

# M Hamza Sajjad

AI/ML Engineer • LLM Systems • Agentic AI • RAG

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## Professional Summary

Artificial Intelligence graduate with three internships spanning machine learning, LLM systems, and multi-agent architectures. Built a production style multi-agent medical AI platform (LangGraph, RAG, Mistral-7B, computer vision) achieving sub-2s response times and 92%+ retrieval relevance, and an EEG-drug simulation model reaching 0.98 ROC-AUC with validated cross site generalization. National Finalist, Huawei ICT Competition. Core stack: Python, PyTorch, FastAPI, LangGraph. Seeking full time AI/ML Engineer roles.

## Projects

### EEG-Guided Digital Brain Twin for Alzheimer's Drug Simulation *PyTorch* | *CVAE* | *ChemBERTa* | *HuggingFace*

- Built a multimodal CVAE fusing EEG spectral features with ChemBERTa drug embeddings to simulate drug conditioned neural responses, achieving ROC-AUC 0.98 and Pearson  $r = 0.867$ , with cross site generalization on external cohorts (OSF, BrainLAT) validated via Integrated Gradients explainability.

### Multi-Agent Medical Assistant System

*LangGraph* | *Qdrant* | *Mistral 7B* | *AWS EC2*

- Architected a production multi-agent pipeline (guardrail, router, RAG, vision, and voice nodes) on AWS EC2 achieving sub-2s response times, 92%+ retrieval relevance, and 60% lower inference cost, with MobileNetV3 classifiers for Brain MRI, Chest X-ray, and Skin Lesion analysis

### LLM-Powered AI Chatbot with Long-Term Memory

*Mistral 7B* | *ChromaDB* | *MLflow*

- Built a locally hosted RAG chatbot with persistent long term memory (ChromaDB cosine retrieval), token level streaming, and a modular LLM abstraction supporting 4 swappable backends (Mistral-7B, HuggingFace, OpenAI, DeepSeek) via single-line config, with MLflow tracking latency and retrieval quality.

### CIRO Crisis Intelligence & Response Orchestrator

*FastAPI* | *React 19* | *Google Gemini* | *LangGraph*

- Designed a 7-agent orchestration pipeline for real time crisis detection, signal fusion, and resource allocation with false positive verification, a live React/Leaflet operations dashboard, and automated multilingual (Urdu/English) stakeholder communication for concurrent geographically distributed emergencies.

## Professional Experience

### Data Analytics & AI Intern

Jul – Aug 2025

*AFT / Digitt++* | *Lahore, Pakistan*

- Built an XGBoost ensemble for fintech fraud detection on 100K+ transaction records using SMOTE and SHAP-based feature selection, achieving a 9% uplift in fraud recall.
- Engineered 5+ automated dashboards (Pandas, Matplotlib, Seaborn), reducing manual reporting effort by 40%; insights informed 3 product feature updates.

### Machine Learning Intern

May – Jun 2025

*Arch Technologies* | *Remote* ()

- Developed Random Forest and gradient boosting classifiers for business analytics (F1 0.88+) using stratified cross validation and L2 regularization on imbalanced datasets

### AI & ML Engineering Intern ★ Best Performance Award

Dec 2024 – Feb 2025

*DevelopersHub Corporation* | *Remote*

- Built machine learning pipelines for classification, regression, and clustering using Scikit-learn and PyTorch, earning Best Performance Award for the internship cohort.

## Education

### B.Sc. in Artificial Intelligence — University of Wah, Pakistan

2022 – 2026 | *CGPA: 3.4/4.0*

Relevant Coursework: Deep Learning, NLP, Computer Vision, Machine Learning, Probability & Statistics, Data Structures & Algorithms, Database Systems

## Core Skills & Technologies

**Languages:** Python, C++

**Machine Learning:** Scikit-learn, XGBoost, Feature Engineering, SMOTE, Cross-Validation, SHAP

**Deep Learning:** PyTorch, CNNs, RNNs, CVAE, Transformers, HuggingFace, Sentence Transformers

**LLMs & GenAI:** RAG, LangChain, LangGraph, Prompt Engineering, Fine-tuning, Mistral 7B, OpenAI API, Google Gemini

**Vector & Data:** ChromaDB, Qdrant, Embeddings, Semantic Search, Pandas, NumPy

**MLOps & Deploy:** FastAPI, Docker, MLflow, AWS EC2, Git/GitHub, Linux

## Certifications & Achievements

- National Finalist : Huawei ICT Competition, Computing Track (2024–2025)
- 1st Place : HITEC University National Project Exhibition (2026): Best AI/Final Year Project "EEG-Guided Digital Brain Twin for Alzheimer's Drug Simulation"
- 1st Place : Open House & Job Fair, University of Wah (2026): Best Final Year Project, Department of Computer Science
- Associate AI Engineer for Developers : DataCamp (Apr 2026, 26 hrs | Credential #837799)